

Practice Problems Sheet V

(Stirling Numbers and Partitions)

Q1. Give direct proof of the following identities for Stirling numbers of the second kind.

(a) $S(n, 2) = 2^{n-1} - 1$

(b) $S(n, n-1) = \binom{n}{2}$

Q2. For the Stirling numbers of the second kind show that

$$S(n, k) = \sum_{r=0}^{n-k} \binom{n-1}{r} S(n-1-r, k-1) = \sum_{r=k-1}^{n-1} \binom{n-1}{r} S(r, k-1)$$

[Hint: Fix an element of n -set and consider all possibilities for the part containing this element.]

Q3

(a) For $n \geq m$, show that $\sum_{k=1}^m S(m, k) n(n-1) \dots (n-k+1) = n^m$.

(b) Does the above equation hold for $n < m$ also? Justify your answer.

Q4. Write all partitions of 7 in the standard form.

Q5. List all self-conjugate partitions of 20 using the result

$$p(n | \text{self conjugate}) = p(n | \text{all parts odd and distinct})$$

proved in class.

Q6. Show that the number of self-conjugate partitions of n with the largest part equal to k is the same as the number of self-conjugate partitions of $n - 2k + 1$ with the largest part not exceeding $k - 1$. Using this find the number of self-conjugate partitions of 41 in which the largest part is 11.

Q7. Show that the number of partitions of n into k parts satisfies $p_k(n) \geq \frac{1}{k!} \binom{n-1}{k-1}$.

—————End—————